

D-James and the Giant Rank Defect

Practical Equivalent Signing-Key Recovery against q5/128

ePrint Signature Audit
AI-assisted cryptanalysis report

7 October 2026

AI provenance and review status. Target screening, attack-history searches, target-specific attack discovery, derivation, implementation, experiments, adversarial review, and the initial manuscript were produced in an AI-run audit by OpenAI Codex and purpose-built specialist AI agents. OpenAI Codex generated and selected the paper title, including its scheme-specific wordplay. Separately instructed agents reproduced the recovery and completed fresh-message signing on two independently generated keys in addition to the primary execution. The technical, adversarial, independent-reproduction, final-history, and publication reviews passed; this report is classified **VERIFIED_ATTACK**.

Abstract

We give a practical public-key-only equivalent signing-key recovery against the advertised q5/128 parameter row of D-James, IACR ePrint 2026/1650, version 5. The public Dragon signature-hash tensor is an unknown-basis expansion of an $\mathbb{F}_5$ -subcode of a two-dimensional generalized Gabidulin code over $\mathbb{F}_{5^4}$. We specialize Vinçotte’s published structural attack against Miranda to recover the hidden field expansion and projected output trace support. We then use the public signature-signature block to decode the HFE coefficients as a dimension-three Gabidulin word with a rank-four error, recover the two-dimensional internal-perturbation plane, complete an equivalent projected HFE-IP map, and recover all Dragon linear maps.

The attack uses only public coefficients. On a sanitized full-parameter public key, independent recovery took 97.62 seconds and 192.25 MB. Completing the equivalent key and signing took 6 minutes 27.45 seconds and 104,960 KiB peak RSS. The recovered signer produced a nonzero 94-symbol signature that satisfies all 73 original public equations for a fresh message and the paper’s all-ones public target. Direct public evaluation rejected the same signature for every one of 256 salts of a changed message. The attack makes no signing queries and does not recover or use the nominal secret key.

No current version-5 implementation is available, and the paper leaves its hash, salt, field, and wire formats unspecified. Our full coefficient instance uses the exact q5 algebra and distribution stated in version 5, generated by a pinned detailed implementation that predates the nonzero right-hand side. We instantiate the new right-hand side with the paper’s all-ones example and use the pinned implementation’s SHAKE256 transcript for the fresh-message demonstration. Equivalent-key recovery signs arbitrary public hash vectors and is independent of those byte-level choices. The result was AI-discovered and independently AI-reproduced, including two additional freshly generated public keys, and passed the project’s technical, history, and publication-review gates.

Result status. Practical full-parameter design attack: public-key-only equivalent signing-key recovery and fresh signing for D-James q5/128. The component Gabidulin-recovery techniques are published work and are credited below. The exact-target identification, completion, and application were not located previously. The result is classified **VERIFIED_ATTACK**.

1 Introduction

D-James [1] combines HFE with minus and internal-perturbation modifiers and adds Dragon bilinear terms between the signature variables and message-hash variables. The Dragon construction permits a hash dimension larger than the number of public equations while keeping the signature short. Version 5 changes verification from a homogeneous equation to a fixed public nonzero right-hand side, blocking the zero-signature forgery that affects the earlier homogeneous formulation.

The same Dragon terms reveal a different structural weakness. After the hidden output map and minus projection, every public hash slice is a base-field expansion of a word in a small generalized Gabidulin code. D-James publishes only a random $\mathbb{F}_q$ -subcode, but recent work on matrix-Gabidulin masking shows how to recover the hidden extension-field structure. Le attacks Enhanced Gabidulin Matrix Codes [3]; Vinçotte develops a particularly direct recovery for the Miranda signature scheme [2]. Random $\mathbb{F}_q$ -subcodes and their residual invariants were also studied by Metouké et al. [4].

Our contribution is to identify D-James’s public Dragon tensor as an instance of this hidden-code problem and to carry the recovery through every remaining public block to a practical signer. The cross tensor recovers the field basis and output trace support but does not itself expose an HFE inverse. The public signature–signature block supplies the missing information. After polarization, its HFE component is a sparse linearized polynomial and its internal-perturbation component is a low-rank error. Rank-metric decoding and base-field linear algebra recover an equivalent central map, including the IP plane and all projected coefficients. The resulting object signs fresh digests against the fixed nonzero target.

Our concrete contributions are:

- an exact trace-adjoint derivation showing that the q5 Dragon tensor is an $\mathbb{F}_5$ -subcode of an expanded two-dimensional generalized Gabidulin code;
- a normalized bivariate determinantal recovery that avoids the large generic Macaulay matrix and recovers the full field basis at $n = 94$;
- a public rank-metric decoder for the q5 HFE coefficients and rank-four IP residual, followed by equivalent IP and Dragon completion;
- a full q5/128 execution producing a fresh accepted signature with zero signing queries, plus public-only and changed-message controls; and
- a sanitized reproducer whose attacker inputs contain no deterministic key-generation seed or secret-derived validation data.

Attack-history statement. We located no previous public attack against exact D-James version 5 in searches as of 7 October 2026. A final search covered the exact scheme name, title, ePrint identifier, authors, source repositories, issue and commit records, and combinations with attack, cryptanalysis, forgery, Gabidulin, matrix code, Miranda, equivalent key, and related terms. We separately searched the three structural papers used here. None names D-James or completes its Dragon, HFE-IP, minus, and fixed-target layers. The negative search has moderate confidence because the target and decisive structural paper are recent. We make no absolute priority claim.

Technique attribution. The hidden matrix-Gabidulin recovery is a specialization of Vinçotte’s attack on Miranda [2], which builds on Le’s EGMC cryptanalysis [3]; we claim no novelty for those primitives. We also claim no novelty for Gabidulin decoding, trace-adjoint representations, or low-rank internal-perturbation analysis. The result here is their exact-target identification and composition into a public-only equivalent signer for D-James.

2 Exact target and scope

2.1 The q5/128 row

The target is D-James in ePrint 2026/1650, version 5, dated 24 September 2026 [1]. The audited PDF has SHA-256 digest

3cf3347c43d37a2b9189d3ab20c44ec12541bd945193c5604efc439224b46f58.

The affected advertised row is

$$(q, n, m, a, n_y, r, D) = (5, 94, 73, 21, 111, 2, 6),$$

with HFE monomials X^2 and $X^6 = X^{q+1}$. Here $a = n - m$ is the number of minus coordinates, n_y is the number of public hash coordinates, r is the number of internal-perturbation forms, and D is the central degree. The paper assigns this row 128-bit classical security and a 219-bit fast signature.

Let $P : \mathbb{F}_5^n \times \mathbb{F}_5^{n_y} \rightarrow \mathbb{F}_5^m$ be the public quadratic/bilinear map. Version 5 verification accepts a signature vector a and public hash vector y exactly when

$$P(a, y) = c,$$

where c is a fixed public nonzero parameter. The paper gives the all-ones vector as its concrete example. Our execution uses $c = \mathbf{1}_m$.

2.2 Available source boundary

The paper-linked C/C++ repository was unavailable during the audit. The author’s Sage notebook and the detailed independent Python/Rust implementation both predate version 5 and use the earlier homogeneous relation. We pin the latter to commit

4328ab366c066554094033297eb1ef3ec8b21c6f.

It supplies a concrete q5 key generator whose coefficient distribution agrees with the literal version-5 algebra: the two table-listed HFE monomials, bilinear IP degree strictly below q^d (hence $k < d$), two IP forms, Dragon maps L_0, L_1 , random secret input/output maps, and the 21-coordinate minus projection. We replace only the public right-hand side with the all-ones example.

The pinned implementation’s field polynomial, SHAKE256 hash-to-field map, 256-salt domain, root order, and encoding are implementation choices because version 5 does not specify them. Our algebraic recovery uses none of these choices except the public coefficient representation. The final transcript uses the pinned SHAKE256 mapping to demonstrate a concrete fresh message.

2.3 Adversarial input hygiene

The initial deterministic experiment record included its generation seed as metadata. Although no attack script read that field, disclosure would permit regenerating the nominal secret and therefore fails a strict public-only review. We removed all seed labels and seed digests, regenerated every derived attacker input from a sanitized public-coefficient file, and reran independent recovery from that file. The sanitized public artifact has SHA-256 digest

51c295dbf1ecd9ce77a890b5436391365f919505971f5a1b19e8f17341dc1e3a.

It contains only public coefficients, dimensions and orientation, the public right-hand side, public messages and their public hash vectors, and source provenance. Secret-derived benchmark fixtures are not part of the attack or public reproducer.

3 The Dragon tensor is a hidden Gabidulin subcode

Set $\mathbb{K} = \mathbb{F}_{q^n}$ and choose an unknown $\mathbb{F}_q$ -basis $\sigma = (\sigma_1, \dots, \sigma_n)$ so that

$$X = \sum_{i=1}^n a_i \sigma_i.$$

Every retained output coordinate is an $\mathbb{F}_q$ -linear functional on $\mathbb{K}$. By nondegeneracy of the trace pairing, it has a unique representation

$$\phi_j(Z) = \text{Tr}_{\mathbb{K}/\mathbb{F}_q}(t_j Z).$$

The minus map retains m independent rows of an invertible output map, so $t_1, \dots, t_m$ are linearly independent over $\mathbb{F}_q$.

For q5, the Dragon contribution associated with hash coordinate s is

$$\Lambda_{0,s}X + \Lambda_{1,s}X^q.$$

Let $\gamma = (\gamma_1, \dots, \gamma_n)$ be trace-dual to σ . If $c_{s,i,j} \in \mathbb{F}_q$ is the public coefficient of $a_i y_s$ in output equation j , define the public slice after unexpansion by

$$d_{s,j} = \sum_{i=1}^n \gamma_i c_{s,i,j}.$$

Using $a_i = \text{Tr}(\gamma_i X)$ and the trace-adjoint identity gives

$$d_{s,j} = t_j \Lambda_{0,s} + (t_j \Lambda_{1,s})^{q^{-1}}.$$

Lemma 1. *For a Dragon central map of q -degree d , all unexpanded public hash slices belong to the generalized Gabidulin parent*

$$\text{span}_{\mathbb{K}}\{t, t^{q^{-1}}, \dots, t^{q^{-d}}\} \subseteq \mathbb{K}^m.$$

The public minus map shortens t but does not destroy the parent as long as the retained entries remain $\mathbb{F}_q$ -independent.

For the target row, $d = 1$. The 111 public slices are therefore a random $\mathbb{F}_5$ -subcode of

$$G_2(t) = \text{span}_{\mathbb{K}}\{t, t^{q^{-1}}\}.$$

The parent has base-field dimension $2n = 188$ and the public subcode has codimension $188 - 111 = 77$. Secret input and output maps change the expansion basis and evaluation vector; they preserve this statement.

4 Normalized public basis recovery

Vinçotte's general recovery reduces the hidden-basis problem to a rank-one extension-field system. The special parent dimension two permits a smaller normalized elimination. Choose three public output columns. From the three entries of each compressed public matrix-code word, form public matrices $A_0, A_1, A_2 \in \mathbb{F}_5^{111 \times 94}$ such that

$$A(u, v) = A_0 + uA_1 + vA_2.$$

On the projective chart whose first compressed coordinate is nonzero, the rank-one equations are equivalent to

$$A(u, v)\gamma^T = 0.$$

Thus the hidden basis occurs on the determinantal locus $\text{rank } A(u, v) < 94$.

We use three 94×94 maximal minors with row sets

$$1:94, \quad 18:111, \quad (1:93) \cup \{95\}.$$

Computing these determinants symbolically was unnecessarily slow. Each minor has total degree at most 94, so we evaluate it on a 95×95 grid over $\mathbb{F}_{5^3}$, interpolate in each variable, verify that the coefficients descend to $\mathbb{F}_5$, and verify the degree bound. The three minors contain 3,704, 3,635, and 3,625 nonzero terms.

Let D_1, D_2, D_3 denote the interpolated minors. We compute

$$R_{12} = \text{Res}_u(D_1, D_2), \quad R_{13} = \text{Res}_u(D_1, D_3).$$

Both resultants have degree 8,836 in v . Their gcd has degree 95 and factorization degrees 1 and 94. Adjoining a root of the unique degree-94 factor defines a recovered copy of $\mathbb{F}_{5^{94}}$; the gcd of all three minors in u is linear. The resulting full pencil has rank 93 and a one-dimensional right kernel. Its kernel entries have base-field rank 94 and therefore form a valid hidden expansion basis.

Unexpanding all 111 public slices gives extension-field rank two. If G is their $\mathbb{K}$ -row space, then

$$\dim_{\mathbb{K}}(G \cap G^{[q]}) = 1.$$

This intersection recovers $\text{span}_{\mathbb{K}}(t)$. The recovered representative t has base-field rank 73, as required by the retained output map.

5 Recovering the HFE-IP block

The cross tensor supplies the hidden input basis and output trace support but not yet a central inverse. We recover the remaining structure from the public aa block.

For output equation j , let Q_j be its public scalar quadratic form in the signature variables. Polarize it and define the trace-adjoint operator $L_j : \mathbb{K} \rightarrow \mathbb{K}$ by

$$Q_j(X + Y) - Q_j(X) - Q_j(Y) = \text{Tr}(YL_j(X)).$$

All coefficients of L_j are computable from the public form and the recovered basis.

The q5 HFE core is

$$\lambda_0 X^2 + \lambda_1 X^{q+1}.$$

Its contribution to the adjoint operator is

$$H_j(X) = 2t_j\lambda_0 X + t_j\lambda_1 X^q + (t_j\lambda_1 X)^{q^{-1}}.$$

It therefore has linearized support $\{-1, 0, 1\}$, or consecutive support of dimension three after a Frobenius shift.

Write the two IP forms as $z_\ell(X) = \text{Tr}(\zeta_\ell X)$. The remaining operator $E_j = L_j - H_j$ has image contained in

$$\text{span}_{\mathbb{F}_q} \{\zeta_1, \zeta_2, t_j\mu_1, t_j\mu_2\},$$

and hence rank at most four. Each known L_j is therefore a sparse dimension-three Gabidulin word plus a rank-four error.

The implementation recovers an image-space annihilator

$$V(Y) = v_0 Y + v_1 Y^q + v_2 Y^{q^2} + v_3 Y^{q^3} + Y^{q^4}$$

from the cyclic recurrence on the known linearized coefficients. It then solves a base-field linear system for the three erased error coefficients. All 73 residual operators have rank exactly four. Their common image-space intersection is

$$Z = \text{span}_{\mathbb{F}_q} \{\zeta_1, \zeta_2\}, \quad \dim_{\mathbb{F}_q} Z = 2.$$

Choose X_1, X_2 trace-dual to any recovered basis of Z . Modulo Z ,

$$E_j(X_\ell) = t_j\mu_\ell \pmod{Z}.$$

Solving these membership constraints for all 73 outputs yields compatible equivalent μ_1, μ_2 . The remaining image-in- Z terms give the three projected coefficients of the quadratic polynomial in (z_1, z_2) . Each is lifted through the surjective trace map

$$C \mapsto (\text{Tr}(t_1 C), \dots, \text{Tr}(t_m C)).$$

The 21 omitted output coordinates create equivalent choices, not an obstacle. Eight independent random public evaluations confirm exact reconstruction of the original projected *aa* map.

6 Dragon completion and fresh signing

For each public hash coordinate s , the recovered cross slice obeys

$$d_{s,j} = t_j \Lambda_{0,s} + t_j^{q^{-1}} \Lambda_{1,s}^{q^{-1}}.$$

Two output positions with an invertible 2×2 coefficient matrix recover $\Lambda_{0,s}, \Lambda_{1,s}$. All 111 pairs fit all 73 public outputs.

For the fixed all-ones right-hand side, choose any lift $C \in \mathbb{K}$ satisfying

$$\text{Tr}(t_j C) = 1 \quad (1 \leq j \leq 73).$$

The trace map has rank 73, so such a lift always exists. More generally, the same step lifts any public m -coordinate right-hand side.

Given a public hash vector y , the equivalent signer computes the recovered Dragon coefficients, enumerates all 25 values of $(z_1, z_2) \in \mathbb{F}_5^2$, and roots the resulting degree-six polynomial over $\mathbb{K}$. It retains a root only if the two recovered IP forms equal the guessed values, then returns

$$a_i = \text{Tr}(\gamma_i X).$$

The full execution found a valid root after 19 degree-six attempts at salt 0 for the previously unsigned message

D-James Miranda transfer fresh message.

The output is a nonzero vector of 94 elements of $\mathbb{F}_5$. Direct evaluation of the original 4,465 public *aa* vectors and 111 public cross slices returns $\mathbf{1}_{73}$. Reusing the same vector for the changed message

D-James Miranda transfer changed message

fails for every salt in the pinned 256-salt domain.

Theorem 1. *For the executed D-James q5/128 public key, the recovered public data form an equivalent signing witness: they reconstruct the entire projected public map and produce a fresh nonzero signature accepted by the ordinary public verification equations for the fixed all-ones target.*

7 Implementation and measurements

All algebra was exact. Public export and verification use Python; field and polynomial recovery use Magma V2.29. Every reported attack run used one worker on an AMD EPYC Processor under KVM. Table 1 records the primary measurements.

Stage	Certificate	Measured cost
Three determinant interpolations	3,704 / 3,635 / 3,625 terms	19.51 / 19.57 / 19.61 s
Two resultants	degree 8,836 each	5.35 / 5.31 s
Independent basis recovery	rank 93, γ rank 94, parent rank 2, intersection 1, t rank 73	97.62 s, 192.25 MB
HFE rank decode	support $\{0, 1, 93\}$; 73 error ranks equal 4	1.52 CPU s after conversion
Post-recovery completion/signing	all 111 Dragon pairs; full public-map certificate; fresh signature	386.92 CPU s; 6:27.45 wall
Post-recovery peak RSS	—	104,960 KiB
Dependency-free public verification	73/73 fresh equations; 0/256 changed salts	0.52 s, 37,376 KiB
Fresh-key executions	2/2 additional keys; 73/73 fresh equations and 0/256 changed salts in each trial	98–99 s recovery plus 391–392 s signing

Table 1: Primary q5/128 measurements.

The generic transferred Miranda model has parent dimension $\kappa = 2$, subcode codimension $\ell_s = 77$, and $\rho = n + \ell_s = 171$. Its degree-three Macaulay column count is

$$\binom{\rho + 2}{3} = 848,046.$$

A dense base-field realization was estimated near 194 GiB. The normalized two-variable determinant route stays below 0.21 GiB and finishes in minutes.

The dependency-free verifier independently recomputes all 512 SHAKE256 hash vectors rather than trusting those stored with the public key. It then evaluates the public equations directly from the sanitized coefficient JSON and claimed signature. A separate Magma recovery regenerated every determinant from the sanitized JSON and loaded no prior determinant, resultant, field modulus, recovered basis, output trace support, key seed, or secret fixture.

Fresh-key reproduction. Two additional full executions started from independently generated, sanitized public keys and imported no primary-key basis or recovery checkpoint. Trial A used the public-input digest

68a358ce0c20fbe7788319d379a6038f79421875a6c690837ccb771aef168401.

Its clean recovery took 98.329 seconds and 224.28 MB. Completion and signing took 391.33 CPU seconds (391.871 seconds wall), found salt 0 after 14 root attempts, and produced a nonzero signature accepted by

all 73 equations. An independent direct verifier recomputed all 512 hash vectors and found no accepting changed-message salt out of 256.

Trial B used the public-input digest

96810a4a5548a6a04aa8e176df12b51e137a6588e475293c5a8d68080ea1da47.

Its three determinant interpolations took 19.55, 19.56, and 19.65 seconds; the resultant stage took 39.37 seconds with 217,088 KiB peak RSS. Completion and signing took 390.80 CPU seconds (391.36 seconds wall), found salt 2 after 63 root attempts, and again gave 73/73 acceptance with zero accepting changed-message salts out of 256. The accompanying `fresh-key-evidence.json` pins the full public-input, recovered-basis, attack-transcript, forgery, and verification hashes for both trials.

Together with the primary execution, all three tested keys succeeded. This small sample demonstrates transfer across independently generated keys; it does not support a precise statistical success-rate estimate.

8 Limitations and claim boundary

Version and wire format. No conforming version-5 implementation was available. We claim the literal paper algebra and its q5/128 coefficient distribution, not conformance to an unpublished byte format. The attack reconstructs an equivalent signer for arbitrary public hash vectors; a future conforming implementation can supply its own hash vector to the same recovered map.

IP range. Version 5 says that the bilinear IP polynomial has degree strictly less than q^d , giving $k < d$ and matching the executed instance. An older author notebook uses $k \leq d$. We do not include that alternate branch in the full-parameter claim.

Other parameter rows. The q5/256 row has the same two-variable parent structure at a larger extension degree, but we did not need to execute it to break an advertised security row. The q2 and q4 rows have parent dimension three and require a multivariate recovery beyond this bivariate implementation. The q13 and q23 rows were not measured. We claim only q5/128.

Equivalent rather than nominal key. Frobenius and scalar equivalences and the output kernel mean that the recovered coefficients need not equal the nominal secret values. This distinction does not reduce impact: the recovered object reconstructs the public map and signs new digests.

Multi-key evidence. The primary deterministic execution and two independently generated fresh-key executions all succeeded. Generality follows from the algebraic derivation; the observed empirical count is only three successes in three executions, and we do not infer a larger statistical success rate from that sample.

9 Conclusion

D-James’s Dragon terms expose a small hidden generalized Gabidulin parent. Published matrix-code recovery techniques reveal its field expansion and output trace support; the public HFE-IP block can then be decoded and completed to an equivalent central map. The complete public-only attack breaks the advertised q5/128 row in minutes and produces a fresh signature against the fixed nonzero target. The nonzero right-hand side prevents the earlier zero forgery but does not protect the structural signing trapdoor.

AI disclosure. Target analysis, history search, derivation, implementation, adversarial review, documentation, and independent reproduction were AI-assisted. Independent AI reproduction is recorded as such. The result is classified `VERIFIED_ATTACK` under the project’s recorded review gates.

References

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